

CHAPTER 01

Activities & Lab Packet

Classifying Life's Diversity

A complete set of labs, worksheets, writing prompts, discussions, and a group project to accompany Chapter 1. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S7L1

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1 Is It Alive? Sorting Lab

LAB

OBJECTIVE

Use the 6 characteristics of life to decide if something is living, once-living, or non-living.

TIME

30 min

STANDARDS

S7L1

Materials (per group of 3–4)

- Sorting-card set or real items: seed, dried leaf, rock, toy car, sponge, yeast packet, apple, salt crystal, feather, photo of a fire
- 3 labeled trays or paper mats: Living / Once-Living / Non-Living
- Student data sheet (next page)

Setup

Set out the items at stations or hand each group a card set. Don't reveal the answers — groups must justify every choice with the 6-trait checklist.

Student Instructions

1. Pick up one item. Run it through the 6 characteristics of life: cells, energy, grow/develop, reproduce, respond, homeostasis.
2. Decide: is it living (does all 6 now), once-living (came from something alive), or non-living?
3. Place it on the matching tray and record your evidence on the data sheet.
4. Be ready to defend a tricky one (seed? yeast? fire?) to the class.

⚠ TEACHER NOTES

Tricky items: A **seed** is living (dormant but can grow). **Yeast** is living (a fungus). A **fire** uses energy and "grows" but isn't made of cells and can't reproduce — non-living. A **dried leaf** is once-living.

Discussion starter: "Why is movement a bad test for life? Name a living thing that barely moves and a non-living thing that does."

ANSWER KEY

Living: seed, yeast, apple (recently), feather-source... | Once-living: dried leaf, sponge (natural), feather | Non-living: rock, toy car, salt crystal, fire. Accept reasoned answers backed by the 6 traits.

Is It Alive? — Data Sheet

Name: _____ Period: _____

Item	Living / Once-Living / Non-Living	Which traits did it show? (evidence)

Conclusion: Why is a single trait (like movement) not enough to decide if something is alive?

2 Mystery Organism — Dichotomous Key

LAB

OBJECTIVE

Use a dichotomous key to identify unknown organisms through either/or questions.

TIME

30–40 min

STANDARDS

S7L1

Materials (per group)

- Set of 6 "mystery organism" picture cards (provided below or teacher-made)
- The dichotomous key on this page
- Data sheet (next page)

The Key — Backyard Mini-Beasts

Step	Choice	Go to / Answer
1a	Has wings	Go to 2
1b	No wings	Go to 3
2a	One pair of wings	Housefly
2b	Two pairs of wings	Go to 4
3a	8 legs	Spider
3b	6 legs	Ant
4a	Colorful, scaly wings	Butterfly
4b	Hard wing covers, spots	Ladybug

Student Instructions

1. Always start at Step 1. Read both choices before deciding.
2. Pick the choice that matches your organism, then follow it to the next step or the name.
3. Keep going until you reach a name — never stop at a question.
4. Record the path you took (1a → 2b → ...) and the final name on your data sheet.

! TEACHER NOTES

Extension: Hand groups a 7th organism not in the key (e.g., a centipede). Ask them to add a new step that would include it. This shows keys can always grow.

Dichotomous Key — Data Sheet

Name: _____ Period: _____

Organism #	Steps I followed (e.g., 1a → 2a)	Identified as...
1		
2		
3		

4		
5		
6		

What does the word "dichotomous" mean, and why does each step have exactly two choices?

4 Levels of Classification

WORKSHEET

OBJECTIVE

Order the eight levels and trace an organism from kingdom to species.

TIME

15–20 min

STANDARDS

S7L1

TEACHER NOTES

The mnemonic from the textbook: **Dear King Philip Came Over For Good Soup** = Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species.

Name: _____ Period: _____

Part A: Put the levels in order from broadest (1) to most specific (8).

Class ____ Species ____ Domain ____ Order ____ Genus ____ Kingdom ____ Family ____ Phylum ____

Part B: Fill in the gray wolf's classification.

Level	Gray Wolf belongs to...
Kingdom	
Phylum	
Class	
Order	
Family	
Genus	
Species	

Part C: Short answer.

- As you move DOWN the levels, do groups get bigger or smaller? Do organisms become more or less alike?

- Which two levels make up an organism's scientific name? _____

ANSWER KEY

Part A: Domain 1, Kingdom 2, Phylum 3, Class 4, Order 5, Family 6, Genus 7, Species 8. Part B: Animalia, Chordata, Mammalia, Carnivora, Canidae, *Canis*, *Canis lupus*. Part C: 1) smaller / more alike; 2) Genus & species.

5 Scientific Naming Practice

WORKSHEET

OBJECTIVE

Apply the rules of binomial nomenclature (two-word naming).

TIME

15 min

STANDARDS

S7L1

The Rules

- Genus first, Capitalized.
- species second, lowercase.
- Whole name *italicized* (or underlined when handwritten).

Name: _____ Period: _____

Part A: Rewrite each name correctly (capitalization + underline).

Written wrong	Written correctly
homo sapiens (human)	
FELIS CATUS (house cat)	
Canis Lupus (gray wolf)	
quercus alba (white oak)	

Part B: True or False.

1. Both parts of a scientific name are capitalized. ____
2. Every species has the same scientific name in every country. ____
3. The genus is the second word. ____
4. Scientific names are usually Latin or Greek. ____

Part C:

Why do scientists use one shared scientific name instead of common names like "robin"?

ANSWER KEY

Part A: *Homo sapiens* | *Felis catus* | *Canis lupus* | *Quercus alba*. Part B: 1) False 2) True 3) False 4) True. Part C: common names differ by language/region, so one shared Latin name prevents confusion.

6 Build Your Own Dichotomous Key

WORKSHEET

OBJECTIVE

Create a working dichotomous key for a small set of organisms.

TIME

20–25 min

STANDARDS

S7L1

Name: _____ Period: _____

Pick 4 organisms (real or imaginary). List one observable trait for each, then write either/or steps that separate them one at a time.

My 4 organisms:

My key:

Step	Choice A → / Choice B →
1a	
1b	
2a	
2b	
3a	
3b	

TEACHER NOTES

A working key for 4 organisms needs **3 either/or steps**. Trade keys with a partner and test whether theirs correctly identifies each organism. If two organisms end at the same place, the key needs another splitting question.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain classification concepts in writing.

TIME

15–30 min each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Alive or Not?** Pick a robot toy. Use the 6 characteristics of life to explain why it is NOT alive, even though it moves and uses energy.
2. **You Are a Taxonomist.** You discover a new organism in your backyard. Describe 3 traits you would look at to decide which kingdom it belongs to.

Level 2 — Intermediate (3 paragraphs)

3. **The Grocery Store.** The textbook compares classification to organizing a grocery store. Extend the comparison: explain how kingdoms, genus, and species are like aisles, shelves, and individual products.
4. **Tools Change Science.** Explain how classification changed as scientists went from using only their eyes → microscopes → DNA. Why does better technology lead to better grouping?

Level 3 — Advanced (essay)

5. **Defend Linnaeus.** Construct an argument that Carolus Linnaeus's naming system was one of biology's most important inventions. Use specific examples of problems it solved.
6. **Why Two Names?** Imagine common names were the only system. Write about three real problems this would cause for scientists around the world, and how binomial nomenclature fixes each.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific reasoning through think-pair-share.

TIME

Warm-ups, exit tickets, seminar

STANDARDS

Cross-cutting

TEACHER NOTES

Use "think-pair-share": 30 sec to think, 1 min with a partner, then call on groups. Several questions have no single right answer — push students to defend with evidence from the chapter.

Quick Warm-Ups (1–2 min)

1. Is a virus alive? Which characteristics of life does it have or lack?
2. Could a fire ever be classified as living? Why or why not?
3. Why is a seed considered alive even when it isn't growing?
4. Why can't we just sort organisms by "does it move"?
5. Why did scientists need to add new kingdoms after the microscope?

Deeper Discussions (5–15 min)

6. Two animals look almost identical but have very different DNA. Should they be in the same group? What should scientists trust more — looks or genes?
7. If you found a brand-new organism, what questions would you ask to place it in a kingdom?
8. Why is it useful that the dichotomous key only ever offers two choices at a time?
9. An alien visits Earth and has never seen our naming system. How would you explain why every species needs a two-word Latin name?
10. The system of classification has changed many times. Does that mean earlier scientists were "wrong," or something else?

9 Create-a-Creature Trading Cards

GROUP

OBJECTIVE

Invent an organism, classify it fully, and build a class field guide.

TIME

2–3 class periods

STANDARDS

S7L1

Project Description

Each student (or pair) invents an original organism and classifies it completely. Finished cards are displayed to build a class-wide "field guide" of imaginary life.

Card Requirements

Front

- Creature name + a drawing
- Its **two-word scientific name** (correctly written)
- The habitat it lives in

Back

- Which **kingdom** + 3 traits that justify it
- All 8 **classification levels**
- One **either/or key question** that separates it from a similar creature
- 3 fun facts

Presentation Day

Each student presents for 1–2 minutes: their creature's kingdom, why it belongs there, and its scientific name. Classmates fill in a "Field Guide Quest" sheet noting one fact per creature.

⚠️ TEACHER NOTES

Differentiation: Struggling students can classify a real organism instead of inventing one. **Grading (suggested):** accuracy of classification 40%, creativity 20%, neatness 20%, presentation 20%.

10 Vocabulary Bingo

REVIEW

OBJECTIVE

Review chapter vocabulary in a low-stress game.

TIME

20–30 min

STANDARDS

S7L1

Setup

Give each student a blank 5×5 grid. Put the word list on the board; students fill squares in any order. Middle square is FREE. Read a clue; students mark the matching word. First to 5 in a row wins.

Vocabulary List (give to students — 24 words)

organism	cell	homeostasis	stimulus	response
classification	taxonomy	kingdom	species	genus
archaeobacteria	eubacteria	protist	fungi	prokaryote
eukaryote	nucleus	dichotomous key	binomial nomenclature	Linnaeus
Aristotle	trait	phylum	characteristics of life	—

Sample Clue Cards (read aloud, don't show)

Clue to read aloud	Answer
A single, complete living thing	organism
The smallest unit of life	cell
Keeping a stable environment inside the body	homeostasis
The science of naming and classifying life	taxonomy
The broadest of the six major groups of life	kingdom
The most specific classification level	species
A cell with NO nucleus	prokaryote
A tool of either/or questions to identify organisms	dichotomous key
The two-word scientific naming system	binomial nomenclature
The scientist who created today's naming system	Linnaeus
A change in the environment that causes a reaction	stimulus
Single-celled organisms that live in extreme places	archaeobacteria

⚠️ TEACHER NOTES

Easy adaptation: Give struggling readers the word list WITH definitions during the game — this turns bingo into low-key matching. Play multiple patterns (line, X, blackout) so several students win.